lecture_08

September 1, 2026

1 Lecture 8: Sensitivity Analysis — Linear Programming

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{note} In Lecture 7, we solved the MTC Chennai bus-service LP by enumerating all
basic feasible solutions (BFS) in a table, and identified the optimal solution
$(x_1^*, x_2^*) = (20, 20)$ with $z^* = 140$ ( '000/day). That answer assumed
the problem data - contribution margins and resource limits - were known with
certainty. In practice, they never are: fuel prices fluctuate, driver-shift
budgets are revised, and planners ask "what if?" before committing to a schedule.
**Sensitivity Analysis** is the systematic method for answering exactly these
questions - without re-solving the LP from scratch.
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Learning Objectives

By the end of this notebook, you will be able to: 1. Extract the inverse basis matrix $\mathbf{B}^{-1}$ directly from the optimal BFS table developed in Lecture 7. 2. Compute shadow prices, reduced costs, resource ranges, and objective ranges using $\mathbf{B}^{-1}$. 3. Interpret sensitivity results to make informed managerial decisions about resource allocation and service planning in transportation contexts.

Prerequisites: LP Formulation (Lecture 3), Graphical Method (Lecture 4), Python SciPy Method (Lecture 6), Simplex Method / BFS Table (Lecture 7)

Estimated time: 90 minutes (including in-class exercises)

1.1 Why Sensitivity Analysis?

Every LP solution rests on a set of parameters: the objective-function coefficients $\mathbf{c}$, the constraint matrix $\mathbf{A}$, and the right-hand-side (RHS) vector $\mathbf{b}$. In real transportation planning, these parameters carry uncertainty:

- The driver-shift budget is set by HR and finance — it can be renegotiated.
- Over what range of driver-shift availability does each additional shift-hour continue to yield the same gain — and when does that rate change?
- The fuel allocation is controlled by the depot manager — it may be expanded for an additional cost.
- The contribution margin of a bus service depends on fare revenue and fuel cost — both volatile.

A decision-maker who takes the LP solution at face value treats these numbers as gospel. A decision-maker who understands sensitivity analysis asks:

> If I secure 5 more driver-shift-hours, how much additional revenue does MTC gain?
>
> Over what range of driver-shift availability does each additional shift-hour continue to yield the same gain — and when does that rate change?
>
> Over what range of fuel budget does the current service mix remain optimal?
>
> By how much can the contribution margin of premium express service fall before we should stop running it at full strength?"

Sensitivity analysis answers all fours questions — and it does so entirely from the optimal BFS table we already computed in Lecture 7.

1.1.1 Shadow Prices

The **shadow price** π_i^* of constraint i is the rate of change of the optimal objective value z^* with respect to a unit increase in b_i, holding all other data fixed:

$$\pi_i^* = \frac{\partial z^*}{\partial b_i}$$

For an LP in standard form, the vector of shadow prices is:

$$\pi^* = \mathbf{c}_B^\top \mathbf{B}^{-1}$$

1.1.2 Reduced Costs

The **reduced cost** $\bar{c}_k$ of variable k measures the net change in the objective per unit increase in $\tilde{x}_k$ from zero, accounting for the fact that increasing $\tilde{x}_k$ forces adjustments to the basic variables to maintain feasibility:

$$\bar{c}_k = c_k - \pi^*[\mathbf{A} \mid \mathbf{I}]_k$$

where $[\mathbf{A} \mid \mathbf{I}]_k$ is the k-th column of the augmented matrix.

For basic variables, the reduced cost is always zero: $\bar{c}_k = 0$ for $k \in$ basis. This is a useful sanity check.

For non-basic variables (currently at zero), the reduced cost indicates:

- $\bar{c}_k < 0$: increasing x_k from zero reduces z — correctly excluded from the basis in a maximization problem.
- $\bar{c}_k = 0$: the problem has alternative optima — another BFS achieves the same z^*.
- $\bar{c}_k > 0$ (maximization): should not occur at optimality — would mean the simplex missed a better solution.

1.1.3 Resource Ranging

While, shadow prices tell us the *rate* at which z^* changes with b_i, they are valid only while the current basis remains optimal. If b_i increases too much, the geometry of the feasible region shifts enough that a different basis becomes optimal — and the shadow price changes. Thus, **resource ranging** determines the range $[b_i^{\min}, b_i^{\max}]$ over which π_i^* remains valid (i.e., the current basis stays both feasible and optimal).

When b_i changes by Δ_i, the new basic variable values are:

$$\mathbf{x}_B(\Delta_i) = \mathbf{x}_B^* + \Delta_i\,\mathbf{B}_i^{-1}$$

where $\mathbf{B}_i^{-1}$ is the i-th column of $\mathbf{B}^{-1}$.

The current basis remains feasible as long as all basic variable values stay non-negative:

$$\mathbf{x}_B^* + \Delta_i\,\mathbf{B}_i^{-1} \geq \mathbf{0}$$

1.1.4 Cost Ranging

Consistent with resource ranging, **cost ranging** asks: how far can c_j shift before the current optimal basis changes? Notably, the current basis remains optimal as long as all reduced costs of non-basic variables remain ≤ 0 (for maximization).

For a basic variable $\tilde{x}_p$ with updated cost coefficient $c_p + \Delta c_p$, the updated reduced cost of a non-basic variable $\tilde{x}_q$ is:

$$\bar{c}_q(\Delta c_p) = \bar{c}_q - \Delta c_p \cdot (\mathbf{B}^{-1}[\mathbf{A}|\mathbf{I}]_q)_r$$

The basis remains optimal as long as $\bar{c}_q(\Delta c_p) \leq 0$ for all non-basic q.

Here, r is the row index of the current optimal tableau in which $\tilde{x}_p$ is basic — i.e., the row such that $\mathbf{x}_B(r) = \tilde{x}_p$. Further, $[\mathbf{A}|\mathbf{I}]_q$ represents the qth column of the augmented matrix, while $(\mathbf{B}^{-1}[\mathbf{A}|\mathbf{I}]_q)_r$ renders the scalar entity in the rth row of $(\mathbf{B}^{-1}[\mathbf{A}|\mathbf{I}]_q)$.

For a non-basic variable $\tilde{x}_q$ with updated cost coefficient $c_q + \Delta c_q$, if the basis $\mathbf{B}$ remains unaffected, then only that variable's own reduced cost shifts:

$$\bar{c}_q(\Delta c_q) = \bar{c}_q + \Delta c_q$$

The basis remains optimal as long as $\bar{c}_q(\Delta c_q) \leq 0$, i.e.,

$$\Delta c_q \leq -\bar{c}_q \quad \Longrightarrow \quad c_q \in \left(-\infty,\ c_q - \bar{c}_q\right]$$

1.2 In-Class Exercise

1.2.1 Exercise 1

Same as Lecture 7 In-Class Exercise 1

The Chennai Metropolitan Transport Corporation (MTC) is evaluating two new express services on a feeder corridor:

Service	Net contribution margin (’000/trip)
E_1 — Premium Express	4
E_2 — Limited-Stop Express	3

Two resources constrain how many trips of each service MTC can run per day:

- Driver-shift hours: both services require 1 driver-shift-hour per trip; 40 driver-shift-hours are available per day.
- Fuel allocation: E_1 (longer route) consumes 2 units of fuel per trip, E_2 consumes 1 unit; the daily fuel budget is 60 units.

Maximize net contribution.

Decision variables: x_1 = daily trips of E_1 (Premium Express), x_2 = daily trips of E_2 (Limited-Stop Express).

Objective:

$$\max_{x_1, x_2} \; z = 4x_1 + 3x_2$$

Subject to:

$$\begin{aligned} x_1 + x_2 &\leq 40 \quad \text{(driver-shift hours)} \\ 2x_1 + x_2 &\leq 60 \quad \text{(fuel budget)} \\ x_1, x_2 &\geq 0 \end{aligned}$$

Standard form (introducing slack variables $s_1, s_2 \geq 0$):

$$\begin{aligned} x_1 + x_2 + s_1 &= 40 \\ 2x_1 + x_2 + s_2 &= 60 \\ x_1, x_2, s_1, s_2 &\geq 0 \end{aligned}$$

Optimal BFS table (from Lecture 7):

Non-Basic (= 0)	Basic	(x_1, x_2)	(s_1, s_2)	Feasible?	$z = 4x_1 + 3x_2$
x_1, x_2	s_1, s_2	$(0, 0)$	$(40, 60)$	Yes	0
x_2, s_2	x_1, s_1	$(30, 0)$	$(10, 0)$	Yes	120
x_2, s_1	x_1, s_2	$(40, 0)$	$(0, -20)$	**No**	—
x_1, s_2	x_2, s_1	$(0, 60)$	$(-20, 0)$	**No**	—
x_1, s_1	x_2, s_2	$(0, 40)$	$(0, 20)$	Yes	120
s_1, s_2	x_1, x_2	$(20, 20)$	$(0, 0)$	Yes	**140**

At the optimal BFS, the **basic variables** are x_1 and x_2 (the non-basics s_1, s_2 are set to zero). The **basis matrix $\mathbf{B}$** is formed by taking the columns of the augmented constraint matrix $[\mathbf{A} \mid \mathbf{I}]$ that correspond to the basic variables.

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad \mathbf{B}^{-1} = \frac{1}{\det(\mathbf{B})} \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$$

since $\det(\mathbf{B}) = (1)(1) - (1)(2) = -1$.

Now then, everything about the optimal solution — and how it changes when parameters shift — can be read off from $\mathbf{B}^{-1}$. Specifically:

Quantity	Formula	Interpretation
Shadow prices	$\pi^* = \mathbf{c}_B^\top \mathbf{B}^{-1}$	Value of relaxing each constraint by one unit
Reduced costs	$\bar{c}_k = c_k - \pi^*[\mathbf{A} \mid \mathbf{I}]_k$	Net cost/benefit of forcing a non-basic variable into the basis
Resource ranging	$\mathbf{B}^{-1}(\mathbf{b} + \Delta\mathbf{b}) \geq \mathbf{0}$	Range over which current basis remains feasible
Cost ranging	$\bar{c}_q(\Delta c_p) =$ $\bar{c}_q - \Delta c_p \cdot (\mathbf{B}^{-1}[\mathbf{A} \mid \mathbf{I}]_q)_r$	Range over which current basis remains optimal

We will now compute each of these systematically.

1.2.2 Shadow Price

$$\pi^* = \mathbf{c}_B^\top \mathbf{B}^{-1} = \begin{bmatrix} 4 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} (-4+6) & (4-3) \end{bmatrix} = \begin{bmatrix} 2 & 1 \end{bmatrix}$$

So $\pi_1^* = 2$ and $\pi_2^* = 1$.

> **Shadow price $\pi_1^* = 2$ (driver-shift hours)**: If MTC could secure one additional driver-shift-hour per day (raising b_1 from 40 to 41), the daily contribution margin would increase by 2,000. The shadow price tells the MTC planner exactly how much it is worth to negotiate with HR for an extra shift.
>
> **Shadow price $\pi_2^* = 1$ (fuel budget)**: One additional unit of fuel per day is worth 1,000 to MTC. If a fuel unit costs less than 1,000 to procure from an emergency supplier, it is profitable to acquire it.

1.2.3 Reduced Cost

For the slack variables s_1 and s_2:

$$\bar{c}_3 = 0 - \pi^*[\mathbf{A} \mid \mathbf{I}]_3 = 0 - \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -2$$

$$\bar{c}_4 = 0 - \pi^*[\mathbf{A} \mid \mathbf{I}]_4 = 0 - \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1$$

For the basic variables x_1 and x_2 (sanity check):

$$\bar{c}_1 = 4 - \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 4 - (2+2) = 0 \checkmark$$

$$\bar{c}_2 = 3 - \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 - (2+1) = 0 \checkmark$$

> **Reduced cost $\bar{c}_{s_1} = -2$:** Each unit of driver-shift slack (each idle driver-shift-hour) costs MTC 2,000 in foregone contribution. Equivalently, MTC is giving up 2,000 for every driver-shift-hour that sits unused. This mirrors $\pi_1^* = 2$ — shadow price and reduced cost of the corresponding slack are equal in magnitude but opposite in sign.
>
> **Reduced cost $\bar{c}_{s_2} = -1$:** Each unused fuel unit costs 1,000 in foregone contribution.

1.2.4 Resource ranging

Ranging for b_1

$$\begin{bmatrix} x_1^*(\Delta_1) \\ x_2^*(\Delta_1) \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix} + \Delta_1 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 20 - \Delta_1 \\ 20 + 2\Delta_1 \end{bmatrix}$$

Requiring both ≥ 0:

- $20 - \Delta_1 \geq 0 \Rightarrow \Delta_1 \leq 20$
- $20 + 2\Delta_1 \geq 0 \Rightarrow \Delta_1 \geq -10$

Thus, with $\Delta_1 \in [-10, 20]$, we have $b_1 \in [30, 60]$.

> The shadow price $\pi_1^* = 2$ is valid for driver-shift budgets between 30 and 60 hours/day. Within this range, every additional shift-hour is worth exactly 2,000 to MTC.

Ranging for b_2

$$\begin{bmatrix} x_1^*(\Delta_2) \\ x_2^*(\Delta_2) \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix} + \Delta_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 20 + \Delta_2 \\ 20 - \Delta_2 \end{bmatrix}$$

Requiring both ≥ 0:

- $20 + \Delta_2 \geq 0 \Rightarrow \Delta_2 \geq -20$
- $20 - \Delta_2 \geq 0 \Rightarrow \Delta_2 \leq 20$

Thus, with $\Delta_2 \in [-20, 20]$, we have $b_2 \in [40, 80]$.

> The shadow price $\pi_2^* = 1$ is valid for fuel budgets between 40 and 80 units/day. Each extra fuel unit yields exactly 1,000 additional contribution within this range.

1.2.5 Cost Ranging

Ranging for c_1 With current reduced costs of $\bar{c}_3 = -2$, $\bar{c}_4 = -1$, when $c_1 \to c_1 + \Delta c_1$, the reduced costs update to:

$$\bar{c}_3(\Delta c_1) = -2 - \Delta c_1 \cdot (\mathbf{B}^{-1}[\mathbf{A} \mid \mathbf{I}]_3)_1 = -2 - \Delta c_1 \cdot (-1) = -2 + \Delta c_1 \leq 0 \Rightarrow \Delta c_1 \leq 2$$

$$\bar{c}_4(\Delta c_1) = -1 - \Delta c_1 \cdot (\mathbf{B}^{-1}[\mathbf{A} \mid \mathbf{I}]_4)_1 = -1 - \Delta c_1 \cdot (1) = -1 - \Delta c_1 \leq 0 \Rightarrow \Delta c_1 \geq -1$$

Thus, with $\Delta c_1 \in [-1, 2]$, we have $c_1 \in [3, 6]$.

> The current service mix $(x_1^*, x_2^*) = (20, 20)$ remains optimal as long as E_1's contribution margin stays between 3,000 and 6,000 per trip. If c_1 falls below 3,000, MTC should shift more trips to E_2.

Ranging for c_2 With current reduced costs of $\bar{c}_3 = -2$, $\bar{c}_4 = -1$, when $c_1 \to c_1 + \Delta c_1$, the reduced costs update to:

$$\bar{c}_3(\Delta c_1) = -2 - \Delta c_1 \cdot (\mathbf{B}^{-1}[\mathbf{A} \mid \mathbf{I}]_3)_2 = -2 - \Delta c_1 \cdot (2) = -2 - 2\Delta c_1 \leq 0 \Rightarrow \Delta c_1 \geq -1$$

$$\bar{c}_4(\Delta c_1) = -1 - \Delta c_1 \cdot (\mathbf{B}^{-1}[\mathbf{A} \mid \mathbf{I}]_4)_2 = -1 - \Delta c_1 \cdot (-1) = -1 + \Delta c_1 \leq 0 \Rightarrow \Delta c_2 \leq 1$$

Thus, with $\Delta c_2 \in [-1, 1]$, we have $c_2 \in [2, 4]$.

> The current service mix remains optimal as long as E_2's margin stays between 2,000 and 4,000 per trip. If c_2 reaches 4,000, the two services become equally attractive and MTC faces alternative optima.

1.3 Take-Away Exercise

For the following take-away exercises, find the optimal solution, identify the basis, compute shadow prices, reduced costs, resource range, and objective range.

1.3.1 Exercise 1

Same as Lecture 7 Take-Away Exercise 2

A Mumbai-based logistics firm dispatches two container types from its port yard each day: 20-ft containers (x_1) and 40-ft containers (x_2), earning net margins of 8,000 and 5,000 per dispatch (in '000: 8 and 5).

Two yard resources constrain daily dispatches:

- Crane-hours: 4 hours/dispatch for 20-ft, 2 hours/dispatch for 40-ft; 60 crane-hours/day available.
- Gate-clearance hours: 2 hours/dispatch for 20-ft, 4 hours/dispatch for 40-ft; 48 gate-clearance-hours/day available.

Maximize net margins.

1.3.2 Exercise 2

A Jaipur Outer Ring Road (ORR) operations centre manages daily tolling and maintenance crew allocation across two shift types:

- x_1: Day-shift crews (06:00–18:00) — net operational value 6,000/crew-day ($c_1 = 6$)
- x_2: Night-shift crews (18:00–06:00) — net operational value 4,000/crew-day ($c_2 = 4$)

Two resource constraints apply each day: - Supervisor availability: 2 supervisors/day-crew + 1 supervisor/night-crew   50 supervisors available - Vehicle allocation: 3 vehicles/day-crew + 2 vehicles/night-crew   90 vehicles available

Minimize operational cost.

1.3.3 Exercise 3

Same as Lecture 7 Take-Away Exercise 1

A Lucknow-based freight operator must decide how many vehicle-trips of three types — Type A (x_1), Type B (x_2), Type C (x_3) — to deploy today, earning net profits of 50, 40, and 30 (in '00/trip) respectively.

Three resources constrain the day's operations:

- Fuel: 3, 1, 2 units/trip for Types A, B, C; budget = 24 units.
- Loading-dock hours: 1, 2, 1 hours/trip; available = 16 hours.
- Driver shifts: 1, 1, 1 shift/trip; available = 10 shifts.

Maximize net profits.

1.4 Circling Back

- **Lecture 4 (Graphical Method)**: Shadow prices correspond geometrically to the slope of the objective function's movement along the optimal vertex. RHS ranging corresponds to how far a constraint line can shift before the optimal vertex moves to a different corner.
- **Lecture 6 (Python SciPy)**: `scipy.optimize.linprog` with `method='highs'` returns shadow prices directly via `res.ineqlin.marginals`. The manual $\mathbf{B}^{-1}$ computation in this lecture explains *where* those numbers come from.
- **Lecture 7 (Simplex Method)**: The optimal BFS table identifies the basis $\mathbf{B}$. Sensitivity analysis is the natural sequel — it extracts maximum managerial value from the same computational artefact.

1.5 Moving Forward

- **Lecture 9 (Duality)**: Shadow prices $\pi^* = \mathbf{c}_B^\top \mathbf{B}^{-1}$ are exactly the optimal solution of the dual LP associated with the primal. Duality theory provides a deeper theoretical foundation for why shadow prices exist and what they represent, and leads to the dual simplex method and complementary slackness conditions.

- **Lecture 10 & 11 (Transportation Engineering Problems)**: The transportation LP has a special structure that allows shadow prices to be interpreted directly as node potentials (supply and demand prices), connecting sensitivity analysis to economic equilibrium concepts.

1.6 Further Reading

- Vanderbei, R.J. (2014). *Linear Programming: Foundations and Extensions* — Chapter 5: Sensitivity and Parametric Analysis.
- Bertsimas, D. and Tsitsiklis, J.N. (1997). *Introduction to Linear Optimization* — Chapter 5: Sensitivity Analysis.
- Winston, W.L. (2004). *Operations Research: Applications and Algorithms* — Chapter 5: Sensitivity Analysis.
- SciPy documentation: `scipy.optimize.linprog` sensitivity outputs — docs.scipy.org